MP1 Design Document - Tiny SNS

Building a demo social networking service with gRPC and Protocol Buffers. Users can login, follow/unfollow other users, list their followers and also enter “timeline” mode to post messages, and see real-time timeline updates from the people they follow. The system has a server (tsd) that manages everything and clients (tsc) that provide the user interface.

Architecture
- Communication: gRPC with bidirectional streaming for real-time messaging
- Data Storage: Text files for message persistence (username.txt)
- Client-Server: Multiple clients connect to single server

Server (tsd.cc)
- **Login**: Create user on login, prevent duplicate login. For invalid username return FAILURE_INVALID_USERNAME and for duplicate login return FAILURE_ALREADY_EXISTS
- **Follow/Unfollow**: Add/Remove target from user’s followers list. Ensure valid user and target are passed in. For invalid user or target return FAILURE_INVALID_USERNAME. If target is not a follower of user return FAILURE_NOT_A_FOLLOWER. For duplicate follow requests return FAILURE_ALREADY_EXISTS.
- **List**: Display followers list of current user.
- **Timeline**: Receive identifier message from user. For this user, collect all posts from their followers. Users' posts are stored on the server at username.txt file. Ensure only posts after follow_time are selected. Sort posts according to post time (descending order). Send at most 20 messages to the user. Enter an infinite loop for broadcasting + receiving messages. On receiving a message from the user (user’s post), persist this message on the server (username.txt) and then forward this message to all of that user’s followers.

Client (tsc.cc)
- **Login**: On starting a user session, create a client stub to connect to the server. Login immediately.
- **ProcessCommand**: interpret command enter in user session. Only allow FOLLOW, UNFOLLOW, LIST and TIMELINE. Reject everything else with FAILURE_INVALID.
- **Follow/Unfollow/List**: Invoke corresponding method on server. Display response.
- **Timeline**: Enter timeline mode (cannot exit once entered). Start a bi-directional stream. Send an identifier to the server, then create two threads - one for reading and a second one for writing messages to the server.
    - Read thread: Display messages as we receive them from the server.
    - Write thread: Send messages to server and user makes posts. The server will broadcast these to the user's followers.